On Reference Signal Estimation from Noisy Speech Using Deep Learning for Intelligibility Estimation

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***Abstract*—We have been studying objective evaluation methods of speech intelligibility. In order to use the full reference method, which can estimate the speech quality with high accuracy, we propose a method to estimate the clean speech, which becomes the reference signal, from the noisy speech by noise removal. We have constructed a Deep Neural Network to eliminate noise and evaluated its noise removal performance.**

Keywords—speech intelligibility, deep learning, noise reduction

I. INTRODUCTION

In voice communication using mobile phones and the Internet, voice quality deteriorates due to various factors. For example, mixing of noise from the surroundings, distortion at the time of data compression, packet loss generated at the time of data transmission, and the like can be cited. In order to design and manage voice communication systems, it is necessary to evaluate and guarantee the quality of the voice including such deteriorations. Currently, the most reliable quality evaluation method is the subjective quality evaluation method in which a person actually using the system listens to and evaluates voice quality. However, because of its nature, the subjective quality evaluation method requires excessive amount of time, labor and cost. Therefore, research on an objective quality evaluation method is being conducted, which measures physical quantities representing the degree of quality deterioration from observation signals and estimates the subjective quality.

The objective quality evaluation method is classified into two broad types. One is the full reference method that estimates the subjective evaluation values using both degraded speech that passed through the evaluation system, and the original speech before degradation. The other is the non-reference method that estimates quality from only degraded speech. Generally, the full reference method using the original speech as the reference signal boasts higher estimation accuracy than the non-reference method.

If the reference signal is not available, the non-reference method, which does not require a reference signal, is used, but the estimation accuracy is lower than the full reference method. Therefore, we propose a method to estimate the original speech before deterioration by removing noise, and use it as a reference signal, as a method to obtain the same estimation accuracy as the full reference method even when there is no reference signal.

In this research, a full reference model is applied to the estimated reference signal and degraded speech to estimate the voice quality. Finally, we aim to estimate the speech intelligibility by utilizing the correlation between voice quality and speech intelligibility. In this paper, we evaluate the performance of the noise removal method using the Deep Neural Network (DNN) as a preliminary stage.

II. PERFORMANCE EVALUATION OF REFERENCE SIGNAL ESTIMATION

A. Sound Source

The sound source used for training is read word speech of the Diagnostic Rhyme Test (DRT) [3]. This list of speech, is a list composed of word-pairs differing only in its initial phoneme, and is divided into six attributes according to phonemic features. The word-pair used in this paper is the pair of words classified into the "sustention" feature [3]. Speech is read words of 20 words in total, 10 word-pairs by 1 female speaker. The sampling frequency is 16 kHz, the number of quantization bits is 16 bits, and the number of channels is monaural. The noise used is JEIDA-NOISE which is a noise database provided by the electronic cooperative [4]. Seven noise types selected from JEIDA - NOISE were used for training and six were used for testing. The breakdown of the noise is summarized in Table 1.

Table 1. Noise used

Training Data	Test Data
1. Exhibition (booth)	8. Exhibition (aisle)
2. Sorting facility	9. Public telephone booth
3. Heavy traffic road	10. Factory
4. Crowd	11. Train (bullet express)
5. Train (local line)	12. Computer room
6. Air conditioner	13. Ducts
7. Elevator halls	

B. Training Conditions

The models were trained using conditions shown in Fig 1. The number of nodes of the input layer and the output layer are 1771 and 161, respectively. Four hidden layers were used, and the number of nodes for each hidden layer is 2000. Logarithmic power spectrogram of speech including noise was used as input data. A frame for which noise is desired to be removed and five frames before and after the frame are taken as one extended input frame. One frame of the logarithmic power spectrogram of clean speech corresponding to the frame for which noise is desired to be removed was used as the supervisory data. The short-time Fourier transform for obtaining the spectrogram was performed with a frame length of 320 samples, a Hann window for the window function, with 50% overlap. Training was carried out under the conditions shown in Table 2 using this model.

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Table 2. Training Conditions

Item	Contents
Preprocessing of training data	Standardization
Hidden layer activation function	ReLU
Output layer activation function	Linear
Dropout	0.5
Loss function	Mean squared error
Learning rate	0.01
Optimizer	Adam

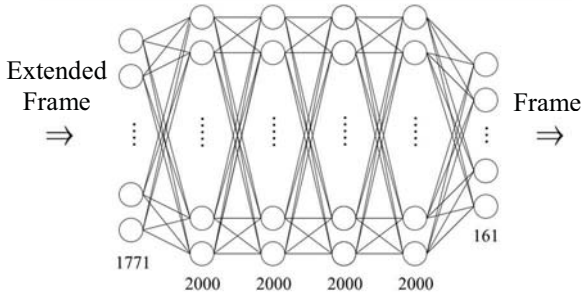


Fig 1. The NN configuration

C. Evaluation Methods

Segmental Signal to Noise Ratio (SSNR) of speech before and after noise removal was used to evaluate noise removal performance. Calculation was performed by the following formula,

$$SNR_{seg} = \frac{10}{L}\sum_{l=0}^{L-1}\log_{10}\frac{\sum_{n=Nl}^{Nl+N-1}x^2(n)}{\sum_{n=Nl}^{Nl+N-}\left(y(n)-x(n)\right)^2} \quad (1)$$

where L is the number of segments, N is the segment length, $x(n)$ is the clean speech, $y(n)$ is the speech before noise removal or after noise removal. In this paper, the segment length N was set to 32 ms. The amount of improvement of SSNR was calculated from the difference before and after noise removal.

III. Evaluation Results

The SSNR before and after noise removal is shown in Fig 2, and the amount of improvement is shown in Fig 3. The improvement amount of SSNR is the difference between SSNR after noise removal and SSNR before noise removal. The horizontal axis is the noise type, and the numbers are the same as those described in Table 1. As can be seen from Fig 3, SSNR is greatly improved with any noise type. In the training data, the average was improved by 17.5 dB and the test data improved by 15.2 dB. However, as can be seen from Fig 2, many SSNRs are low even after removing noise, 0.747 dB on average for all noise types. Especially, noise types with low SSNR before noise removal, noise numbers 1, 2, 12, 13 etc in Fig 2, resulted in lower SSNR after noise removal than other noise types. It seems that the value of SSNR before noise removal also affects other noise even after noise removal. In addition, although the noise 9 is test data and not used for training, the SSNR was the highest among all the noises. This seems to be because the noise 9 is closer to stationary noise compared to other noises. From this fact, it is considered that the stationarity of the noise is closely related to the noise removal performance.

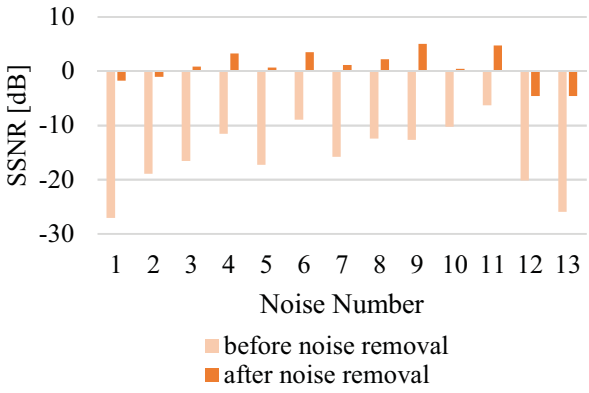


Fig 2. SSNR before and after noise removal

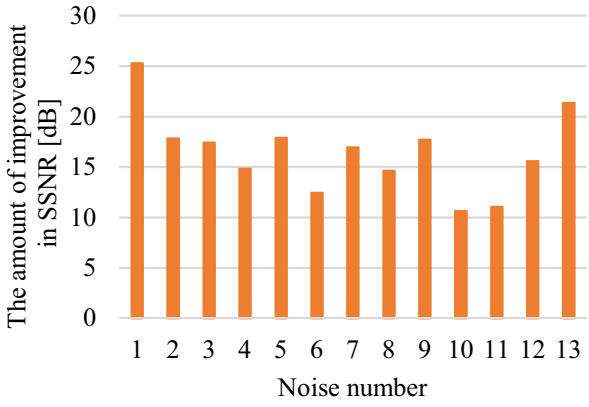


Fig 3. SSNR improvement after noise removal

IV. Conclusion

In this research, we proposed a method for noise elimination using DNN for speech intelligibility estimation, and evaluated its noise removal performance. As a result, the SSNR before and after the noise elimination improved by an average of 17.5 [dB] in the training data and by 15.2 dB in the test data. However, the actual SSNR was 0.747 dB on average, which was still relatively low.

As a future work, we need to optimize the DNN model and improve the noise removal performance. In addition, we would like to estimate intelligibility with the current model, and evaluate to what extent the intelligibility estimation accuracy can be obtained.

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ACKNOWLEDGMENT

This work was supported in part by the JSPS KAKENHI 17K00223.

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